

TIANJIN, China

WMO: 545270

Lat: 39.0812N

Lon: 117.0525E

Elev: 16

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.1	16.2	-13.8	2.6	23.4	-9.6	3.3	25.1	24.5	26.7	21.9	27.1	4.9	0	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	94.7	73.8	92.2	73.9	89.9	73.5	81.6	88.4	80.0	86.4	78.5	84.7	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	154.5	86.1	78.2	146.4	84.6	76.5	138.5	83.2	45.6	88.9	43.8	86.5	42.1	84.8	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	16.7	14.0	8.9	99.2	4.1	2.9	6.0	101.2	3.6	102.9	1.3	104.5	-1.7	106.6		
			6.7	83.3	3.9	1.5	3.9	84.4	1.7	85.3	-0.5	86.2	-3.3	87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	27.1	33.2	45.8	59.2	70.1	77.6	81.5	79.5	71.2	58.4	42.7	30.7
	DBStd	20.02	5.44	6.16	7.49	7.16	5.89	4.92	4.06	4.02	5.42	6.37	7.45	5.16
	HDD50	2208	711	470	174	9	0	0	0	0	0	10	237	597
	HDD65	4841	1176	890	595	200	20	0	0	0	12	218	668	1062
	CDD50	4603	0	0	45	286	625	829	978	914	636	271	19	0
	CDD65	1760	0	0	1	27	179	379	513	449	198	14	0	0
	CDH74	17027	0	0	12	273	1610	3728	5646	4356	1327	75	0	0
	CDH80	6518	0	0	2	58	557	1570	2388	1591	343	9	0	0
(14) Wind	WSAvg	5.8	5.5	5.9	7.2	7.7	7.1	5.9	5.1	4.5	4.7	5.1	5.1	5.3
(15) Precipitation	PrecAvg	20.4	0.1	0.2	0.3	0.9	1.4	3.1	6.1	4.6	1.9	1.1	0.6	0.1
	PrecMax	29.9	0.5	0.7	1.7	3.1	2.8	6.2	12.5	13.5	4.8	5.7	2.2	0.4
	PrecMin	14.3	0.0	0.0	0.0	0.1	0.2	0.2	2.8	1.8	0.1	0.0	0.0	0.0
	PrecStd	4.4	0.1	0.2	0.4	0.9	0.8	1.5	2.2	2.4	1.2	1.1	0.6	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	56.4	74.4	85.8	94.6	98.1	98.1	94.5	89.8	80.4	65.8	48.6
		MCWB	35.1	42.0	53.9	61.2	65.9	69.5	76.7	77.8	70.8	62.2	50.7	37.4
	2%	DB	41.7	51.6	68.2	80.6	89.8	94.5	95.0	91.7	86.4	75.7	61.4	44.8
		MCWB	32.5	39.9	50.2	59.2	64.6	70.1	77.2	76.0	69.4	59.9	48.9	35.2
	5%	DB	38.8	47.8	63.1	76.5	86.0	91.4	92.3	89.5	83.9	72.1	57.6	41.6
		MCWB	30.7	37.6	47.9	57.1	64.3	70.4	76.9	75.9	68.9	58.7	46.7	33.7
	10%	DB	36.0	44.2	58.5	72.7	82.6	88.2	89.9	87.4	81.3	68.9	53.9	38.9
		MCWB	29.1	35.2	45.1	55.3	62.4	69.8	76.4	75.4	67.3	57.0	44.9	32.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	44.1	56.5	65.6	72.2	77.8	83.7	83.2	78.4	66.8	54.0	40.1
		MCDB	43.5	52.2	70.4	79.4	85.4	88.0	91.8	89.2	84.6	73.6	61.3	45.7
	2%	WB	33.6	41.2	52.7	62.7	69.6	75.9	82.2	81.4	74.5	64.1	50.9	37.0
		MCDB	39.6	48.9	64.5	74.7	82.3	86.6	89.7	87.4	80.4	70.4	57.6	41.6
	5%	WB	31.5	38.6	49.3	60.3	67.7	74.3	80.7	79.8	72.4	61.9	48.6	35.0
		MCDB	37.0	46.4	60.5	72.3	80.0	84.2	87.5	85.6	79.0	68.5	55.1	39.7
	10%	WB	30.0	36.1	46.2	57.8	65.7	72.7	79.2	78.3	70.4	59.6	46.4	33.2
		MCDB	35.1	43.1	57.0	69.3	77.5	82.3	85.8	83.8	77.7	66.7	52.8	37.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.6	17.7	19.2	19.3	17.1	13.6	13.6	16.3	17.1	15.3	14.2
		MCDBR	19.1	22.1	24.2	25.4	25.6	22.4	17.8	16.3	19.2	20.8	20.4	18.5
		MCWBR	12.2	12.9	11.2	9.9	8.5	6.8	5.9	5.8	6.4	8.6	10.6	11.4
	5% WB	MCDBR	17.3	20.4	22.7	21.8	20.6	16.6	13.7	12.7	14.6	16.3	17.9	16.0
		MCWBR	11.5	12.9	11.3	10.6	9.1	7.0	6.3	6.2	7.1	8.2	10.4	10.8
(40) Clear-Sky Solar Irradiance	taub		0.469	0.546	0.565	0.629	0.674	0.808	0.807	0.760	0.685	0.654	0.535	0.484
	taud		1.860	1.729	1.715	1.637	1.587	1.417	1.460	1.524	1.598	1.611	1.791	1.878
	Ebn at Noon		203	204	219	215	210	183	181	185	187	172	180	186
	Edh at Noon		47	61	69	79	84	100	95	87	76	68	50	43
(44) All-Sky Solar Radiation	RadAvg		750	973	1414	1688	1885	1682	1516	1472	1315	1010	767	669
	RadStd		75	116	126	91	130	160	108	90	107	102	109	75

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-4.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+110

Nomenclature: See separate page